Table 1. Final multiple linear regression models for the scores

	HVLT-R Total Recall	HVLT-R Delayed Recall	HVLT-R Recognition	ROFC copy	ROCF memory	Letter F	Letter A	Letter S	Letter M	Animals	Fruits
Intercept	19.075	6.203	10.747	28.004	13.085	9.702	9.868	8.898	10.553	15.225	13.637
Age	-0.122	-0.06	-0.025	-0.223	-0.204	-0.061	-0.069	-0.046	-0.085	-0.107	-0.081
Age^2	-0.001	-0.001	0.001	-0.003	4.158	-0.001	-0.001		-0.001	-0.001	-0.001
Education	3.042	1.322	0.356	4.013		3.454	3.932	3.409	4.041	3.951	2.568
Sex	-0.801	--	--	--	1.768	--	--	--	--	--	-1.511
Age x Edu	--	--	--	0.06	--	--	--	--	--	--	--
R^2	0.308	0.253	0.082	0.276	0.356	0.202	0.254	0.192	0.281	0.316	0.27

Table 1. Continue… Final multiple linear regression models for the scores

	SDMT	TMT-A	TMT-B	Boston Naming test	Stroop Word	Stroop Color	Stroop Word-Color	Stroop Interference	M-WCST Categories	M-WCST Perseverations	M-WCST Total Errors
Intercept	26.256	74.158	184.949	39.811	78.951	57.228	31.486	-1.545	3.412	8.404	17.434
Age	-0.399	0.642	2.403	-0.128	-0.690	-0.594	-0.411	-0.089	-0.029	0.086	0.129
Age^2	0.001	0.006	0.009	--	-0.011	-0.009	-0.004	--	--	--	--
Education	13.549	-18.417	-81.197	6.328	13.413	8.393	5.241	--	1.121	-3.436	-5.74
Sex		--	--	--	--	--	1.915	2.855	--	--	--
Age x Edu	0.039	--	--	--	--	--	--	--	--	--	--
Age^2 x Edu	0.002	--	0.035	--	--	--	--	--	--	--	--
R^2	0.519	0.406	0.437	0.213	0.340	0.366	0.361	0.069	0.200	0.112	0.156

Table 2. Standard Deviation (residual) for final multiple linear regression models

Scores	Predicted value	SD (Residual)	Converter Z-scores to Percentile
HVLT-R Total Recall	All values	4.230	Standard normal cumulative distribution function
HVLT-R Delayed Recall	All values	2.262	Standard normal cumulative distribution function
HVLT-R Recognition	≤10.279	2.140	Empirical cumulative distribution function
	10.280 to 10.751	1.700	
	10.752 to 11.108	1.292	
	>11.109	1.228	
ROFC copy	≤24.064	8.566	Empirical cumulative distribution function
	24.065 to 28.267	7.433	
	28.268 to 31.713	6.431	
	>31.714	4.255	
ROCF Memory	≤11.012	5.693	Standard normal cumulative distribution function
	11.013 to 14.484	6.732	
	14.485 to 18.030	7.314	
	18.031	6.752	
Letter A	All values	3.974	Empirical cumulative distribution function
Letter S	All values	4.023	Standard normal cumulative distribution function
Letter M	All values	4.065	Standard normal cumulative distribution function
Animals	All values	4.192	Standard normal cumulative distribution function
Fruits	All values	3.386	Standard normal cumulative distribution function
SDMT	≤20.235	9.970	Standard normal cumulative distribution function
	20.236 to 27.079	10.467	
	27.08 to 37.126	11.563	
	>37.127	11.120	
TMT – A	≤60.384	17.695	Empirical cumulative distribution function
	60.385 to72.774	20.957	
	72.775 to 84.980	20.195	
	>84.981	15.882	
TMT - B	≤122.117	37.717	Standard normal cumulative distribution function
	122.118 to 179.732	69.006	
	179.733 to 222.488	74.558	
	>222.489	69.163	
Boston Naming test	≤38.039	8.856	Empirical cumulative distribution function
	38.040 to 40.607	8.686	
	40.608 to 44.587	8.348	

	>44.588	6.454	
Stroop Total Word	≤25.108	9.677	Standard normal cumulative distribution function
	25.109 to 32.343	10.234	
	32.344 to 38.723	12.358	
	>38.724	11.844	
Stroop Total Color	All values	15.205	Standard normal cumulative distribution function
Stroop Total Word-Color	≤25.108	9.677	Empirical cumulative distribution function
	25.109 to 32.343	10.234	
	32.344 to 38.723	12.358	
	>38.724	11.844	
Stroop Interference	All values	8.431	Standard normal cumulative distribution function
M-WCST Categories	≤3.008	1.801	Empirical cumulative distribution function
	3.009 to 3.593	1.717	
	3.594 to 4.304	1.835	
	>4.305	1.434	
M-WCST Perseverations	≤5.731	4.858	Empirical cumulative distribution function
	5.732 to 7.958	6.396	
	7.959 to 9.671	7.899	
	>9.672	8.811	
M-WCST Total Errors	≤13.083	8.369	Empirical cumulative distribution function
	13.084 to 16.763	9.594	
	16.764 to 19.339	9.788	
	>19.340	9.646	

Normative Procedure

Norms that allow for converting 'raw' test scores into percentile scores (thereby accounting for the significant demographic variables) for the different scores by Colombia were established using the four-step procedure described in the next section:

Using the final regression model that is obtained at the end of the stepwise procedure, normative data that are adjusted for demographic variables were established by means of a four-step procedure (Rivera & Arango-Lasprilla, 2017; Van Breukelen & Vlaeyen, 2005; Van der Elst et al., 2006a; Van der Elst et al., 2006b): 1. The expected test score ($\hat{Y}_i$) is computed based on the fixed effect parameter estimates of the established final regression model: $\hat{Y}_i = B_0 + B_1X_{1i} + B_2X_{2\mathrm{i}} + \cdots + B_{\mathrm{K}}X_{\mathrm{Ki}}$. 2. To obtain the residual value (e_i), a subtraction between the raw score of the neuropsychological test (Y_i) and the predicted value previously calculated was performed ($\hat{Y}_i$), as shown in the following formula: $e_i = Y_i - \hat{Y}_i$. 3. Using the residual standard deviation (SD_e) value provided by the regression model, residuals were standardized: $z_i = e_i / SD_e$. 4. Finally, the exact percentile corresponding to the *z*-score previously calculated was obtained using the standard normal cumulative distribution function (if the model assumption of normality of the residuals was met in the normative sample), or via the empirical cumulative distribution function of the standardized residuals (if the standardized residuals were not normally distributed in the normative sample).

An example is provided below to facilitate the understanding of the procedure used to obtain the percentile value associated with a raw score on this test. To find the percentile score for a 30-year-old Colombian woman who has 13 years of education, and who scored 15 on the fruits category, the steps are as follows:

1) Find in Table 1, which provides the final regression models by country for the fruits category scores. Use the B regression weights to create an equation to predict the

expected fruits category score for a 30-year-old woman with 13 years of education using the coding provided in the statistical analysis section. The corresponding B regression weights are multiplied by the centered age (= calendar age – 58 years), centered age^2 (= calendar age – 58 years)^2. Education was dummy coded into a variable of 0 and 1: 1 if the participant had > 12 years of education and 0 if the participants had 1-12 years of education, and sex was coded as 1= male, and 0=female. Then the result is added to the constant generated by the model in order to calculate the predicted value. Two-level interactions were not significant predictors, and therefore are not included in this model.

In this example of the Colombian woman, the predicted fruits category score can be calculated using the following equation: $\hat{y_i} = 13.637 + [-0.081 \cdot (Age_i - 58)] + [(-0.001 \cdot (Age_i - 58)^2] + [2.568 \cdot Education] + (-1.511 \cdot Sex\ )$. The woman's age is 30 and his education is 13 years. Thus, the predicted value equation is: $\hat{y_i} = 13.637 + [-0.081 \cdot (30 - 58)] + [(-0.001 \cdot (30 - 58)^2] + [2.568 \cdot 1] + (-1.511 \cdot 0\ ) = 17.689$.

2) To calculate the residual value (e_i), subtract the actually observed fruits category score (she scored 15) from the predicted value we just calculated ($e_i = y_i - \hat{y_i}$). In this case, the residual would be $e_i = 15 - 17.689 = -2.689$.

3) Next, consult the SD_e column in Table 2 to obtain the score-specific SD_e (residual) value. Thus, its corresponding SD_e is 3.386. Then, the following formula must be applied: $z_i = e_i\ /\ SD_e = 2.689/3.386 = -0.794$.

4) The last step, standardized residuals value (z) to convert z score to percentile using a normal distribution function. In this example, the z score of -0.794 corresponds to the 21^h percentile. It is important to remember to use the appropriate tables that correspond to each when performing these calculations.